Topic 2 Entropy, Large Deviations, and Free Energy

Review of Last lecture

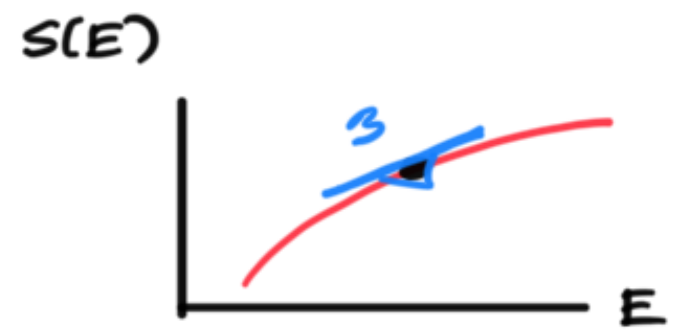
- **closed systems:**

energy is conserved at a fixed value E

$$\boxed{p(\vec{s}) = \frac{1}{\Omega(E)}\,\delta\left[E(\vec{s}) - E\right]}$$ ← distribution over microstates is uniform over all states w/ energy E

$\Omega(E)\,dE$ = # microstates $\vec{s}$ with energy $E(\vec{s})$ in E to $E+dE$

$$\boxed{S(E) = \ln \Omega(E)}$$ microcanonical entropy



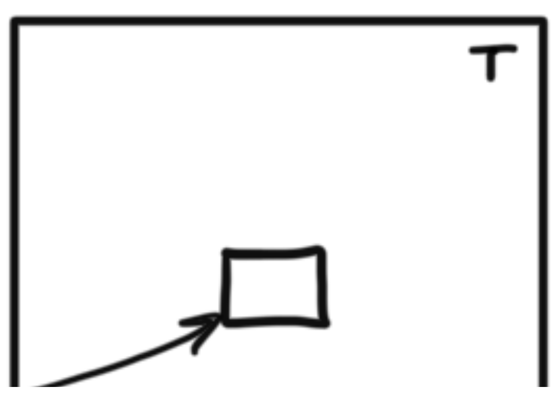
- **temperature:**

$$\boxed{\beta \equiv \frac{1}{kT} = \frac{\partial S}{\partial E}}$$

- entropy maximization ⇒ 2 systems in thermal equilibrium iff $\frac{\partial S_1}{\partial E_1} = \frac{\partial S_2}{\partial E_2}$

or $T_1 = T_2$

- **boltzmann**



← thermal reservoir at temp T

System in eq at temp T

$$\boxed{p(\vec{s}) = \frac{e^{-\beta E(\vec{s})}}{Z}}$$

Boltzmann distribution

$$\boxed{Z(\beta) = \int d\vec{s}\, e^{-\beta E(\vec{s})}}$$

Partition Function

$$\boxed{F(\beta) = -\frac{1}{\beta} \ln Z(\beta)}$$

Free energy

Key intuition:

- microstates $\vec{s}$ of higher energy $E(\vec{s})$
- are exponentially more unlikely
- because such energy must be taken from the reservoir
- which then has exponentially fewer accessible states at its lower energy

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- Now that we have a physical derivation of the Boltzmann distribution we are going to study it from different perspectives

- First : how do we generalize entropy to nonuniform distributions

let $p_i \geq 0$ for $i = 1 \ldots \Omega$ with $p_1 + \cdots + p_\Omega = 1$

denote a nonuniform probability distribution over Ω possible outcomes

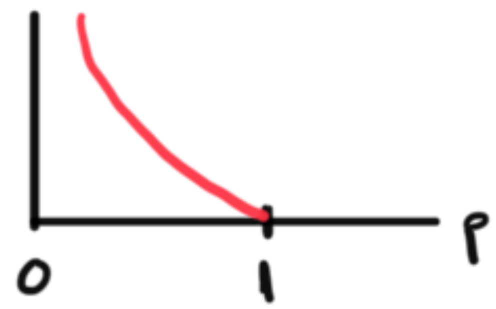
$$S(\vec{s}) = -\sum^{\Omega} p_i \ln p_i$$

Definition of entropy

$$S(p) = \sum_{i=1}$$

Intuition: Entropy as average "surprise"

$-\ln p = \ln 1/p$



low probability events are more surprising

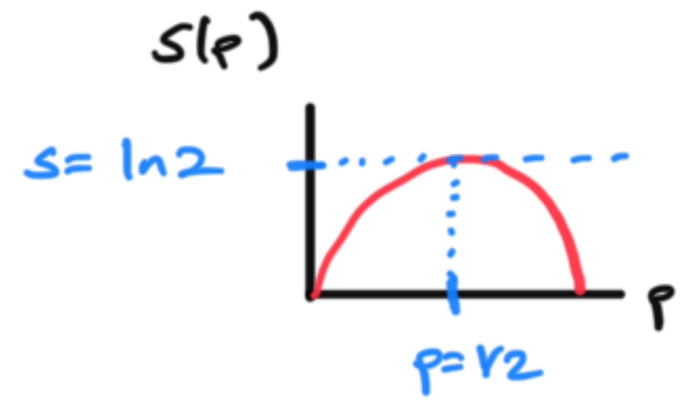
Consistency check: for uniform distributions $p_i = \frac{1}{\Omega}$ for $i = 1 \dots \Omega$

$$S(\vec{p}) = -\sum_{i=1}^{\Omega} \frac{1}{\Omega} \ln \frac{1}{\Omega} = \ln \Omega \quad ✓✓$$

Special case: entropy of biased coin

$\Omega = 2$ $\quad p_1 = p$, $\quad p_2 = 1-p$

$$S = -p \ln p - (1-p) \ln (1-p)$$



Calculation: entropy as logarithm of # of long messages one can construct

- suppose we are given N letters from an alphabet of size K
- but we are given an uneven # of copies of each letter

- let N_1 = # of copies of letter 1

  N_2 = # " " " 2

  $\vdots$

  N_Ω = # of copies of letter K

- Then $p_i \equiv \dfrac{N_i}{N}$ = fraction of letters that correspond to letter i

- how many unique messages (or letter sequences) can we construct? Ω

$$\Omega = \frac{N!}{N_1!\ N_2! \cdots\cdots N_K!}$$

- lets take the asymptotic limit

  $N \to \infty$

  $N_i \to \infty$

  $p_i = \dfrac{N_i}{N} \sim O(1)$

- use Stirlings approximation

$$\boxed{\ln n! \approx n \ln n - n}$$

for large n

$$\ln \Omega = \ln N! - \sum_{i=1}^{K} \ln N_i!$$

$$= N \ln N - N - \sum_{i=1}^{K} \left[N_i \ln N_i - N_i \right]$$

$$= N \ln N - N - \sum_{i=1}^{K} N_i \ln N_i + \underbrace{\sum_{i=1}^{K} N_i}_{= N}$$

Cancel

$$= \sum_{i=1}^{K} N_i \left[\ln N - \ln N_i \right]$$

$$= \sum_{i=1}^{K} N_i \ln \frac{N}{N_i}$$

$$= - N \left(\sum_{i=1}^{K} \frac{N_i}{N} \ln \frac{N_i}{N} \right)$$

$$= - N \sum_{i=1}^{K} p_i \ln p_i$$

$$\ln \Omega = N\, S(\vec{p}) \Rightarrow \boxed{\underbrace{\Omega}_{\text{\# of messages one can construct}} = \underbrace{e^{N S(\vec{p})}}_{\text{grows exponentially with sequence length } N}}$$

exponential rate of growth = entropy $S(\vec{p})$ of source

<u>Dive deeper: Statistical mechanics of independent sequences</u>:
- KL divergence
- Large deviations
- Free energy

- consider a uniform distribution over K outcomes / letters ie $p_i = \frac{1}{K}$ for $i = 1 \dots K$

- consider a long sequence of length N of such letters, each drawn independently

example: $K=3$: 1 1 1 1 1 1 1 1 1 1 prob $= (1/K)^N$

$N=10$: 1 3 2 2 1 1 3 2 3 1 prob $= (1/K)^N$

- these two sequences have the same probability

(in fact there are K^N sequences each with prob $(1/K)^N$)

- but the first sequence seems much more unlikely. why??

- answer: the empirical distribution of the first sequence is very different from the true distribution ...

can think of this as the difference between:

microstates : individual sequences

macrostate : distribution of letters in the sequence

Microstates $\Bigg\{$ let $\vec{s} = (s_1, \dots\dots, s_N)$ sequence of N letters / degrees of freedom

$s_i \in \{1, \dots, K\}$

Distribution over microstates

- $\text{Prob}[\, s_i = j \,] \equiv p_j$ for $j = 1 .. k$ and all $i = 1 .. N$

  ie prob that ith letter in sequence takes the value j is p_j

  $$p_j \geq 0 \qquad \sum_{j=1}^{k} p_j = 1$$

- prob of each letter is indep of every other letter

  $$P(\vec{s}) = \prod_{i=1}^{N} P(s_i)$$

Macrostate

$$\boxed{\hat{p}_j(\vec{s}) = \frac{1}{N} \sum_{i=1}^{N} \delta_{s_i, j}} \qquad \delta_{a,b} = \begin{cases} 1 & \text{if } a = b \\ 0 & \text{otherwise} \end{cases}$$

$\hat{p}_j$ = empirical fraction of times the N letters in sequence $\vec{s}$ take the value j

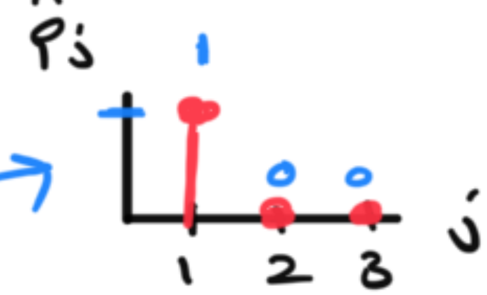
$\hat{p}_j(\vec{s})$ obeys $\hat{p}_j \geq 0$ and $\sum_{j=1}^{k} \hat{p}_j = 1$ so it is a probability distribution

but it is a random probability distribution due to randomness in $\vec{s}$

Example:

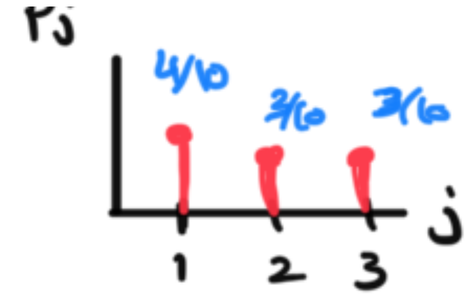
$N = 10$ $\quad \vec{s} = 1\ 1\ 1\ 1\ 1\ 1\ 1\ 1\ 1\ 1$

$k = 3$ $\quad \vec{s} = 1\ 3\ 2\ 2\ 1\ 1\ 3\ 2\ 3\ 1$



Macrostates

Microstates



- the first sequence seems less likely because its macrostate is less likely

Key question

> if I draw an i.i.d sequence $\vec{s}$ from the true distribution p
> What is the probability that its empirical distribution is $\hat{p}$?

$$\boxed{\mathrm{Prob}\left[\,\hat{p}(\vec{s}) = \hat{p}\,\right] = \underbrace{\left[\frac{N!}{\prod_{j=1}^{K}(N\hat{p}_j)!}\right]}_{\text{an "entropic" term}}\underbrace{\left[\prod_{j=1}^{K} \underbrace{p_j}_{\text{true prob of letter } j}^{\overbrace{N\hat{p}_j}^{\#\text{times letter } j \text{ appears in } \vec{s}}}\right]}}$$

an "entropic" term:

# of sequences $\vec{s}$ with empirical distribution $\hat{p}(\vec{s}) = \hat{p}$

we already know that as $N \to \infty$ this is $e^{NS(\hat{p})}$

prob of a i.i.d sequence $\vec{s}$ depends on $\vec{s}$ only through its empirical distribution $\hat{p}(\vec{s})$

- now take large N limit via Stirling's approximation

$$\log \text{Prob}\left[\hat{p}(\vec{s}) = \hat{p}\right] = \underbrace{N\, S(\hat{p})}_{\text{we know this from above}} + \underbrace{N \sum_{j=1}^{K} \hat{p}_j \ln p_j}$$

$$= N\left[-\sum_{j=1}^{K} \hat{p}_j \ln \hat{p}_j + \sum_{j=1}^{K} \hat{p}_j \ln p_j\right]$$

$$= -N\left[\sum_{j=1}^{K} \hat{p}_j \ln \frac{\hat{p}_j}{p_j}\right]$$

$$= -N\, D_{KL}\left[\hat{p}_j \,\|\, p_j\right]$$

$$\boxed{\text{Prob}\left[\hat{p}(s) = \hat{p}\right] \approx e^{-N D_{KL}[\hat{p} \,\|\, p]}}$$

Kullback Leibler divergence

$$\boxed{D_{KL}[p \,\|\, q] = \sum_{j=1}^{K} p_j \ln p_j/q_j}$$

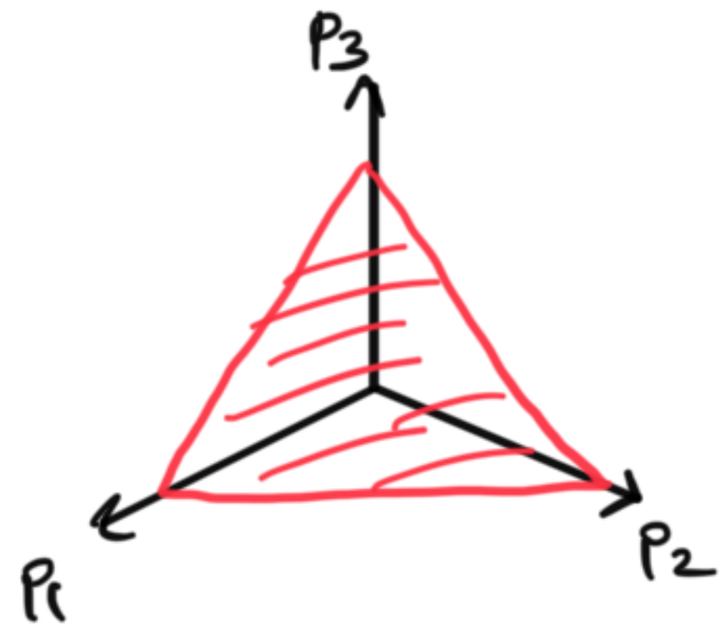
- if true probability distribution of letters is p
- the probability of observing an empirical distribution $\hat{p} \neq p$
- is exponentially suppressed in sequence length N
- unless $\hat{p} = p$ (or $D_{KL}[\hat{p} \,\|\, p] \sim \Theta(1/N)$)

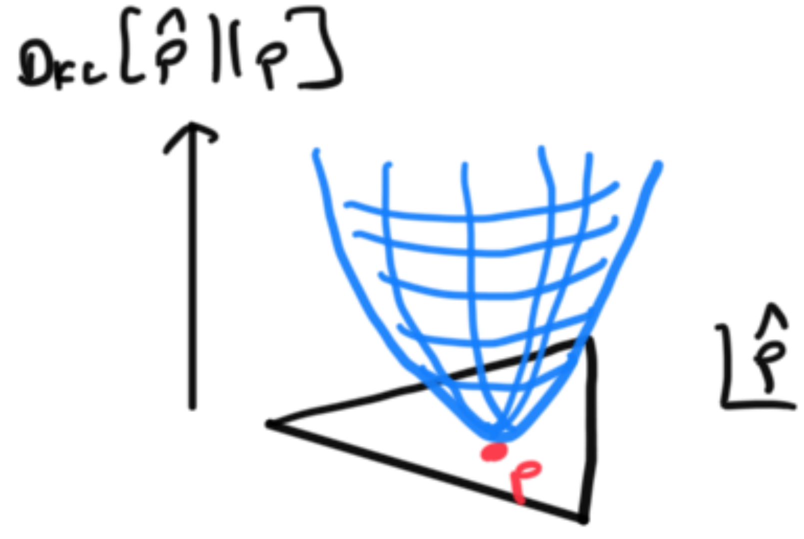
Properties

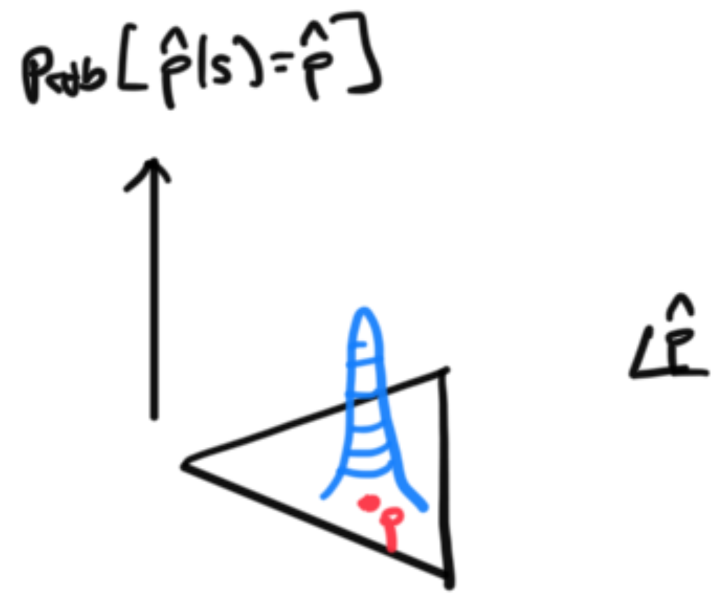
- jointly convex in p and q
- 0 if $p = q$
- $\Rightarrow \geq 0 \;\forall\; p$ and q
- not symmetric

- think of it as a divergence between two distributions

Picture for $K=3$ $\quad p_1+p_2+p_3=1 \quad p_i \geq 0$







Notion of typical and atypical sequences

- informally: a sequence $\vec{s}$ is <u>typical</u> <u>iff</u> its empirical distribution $\hat{p}(s)$ is close to the true distribution p
- in the distribution $p(\vec{s})$ the typical sequences carry all the probability close to 1 for large N
- there are $\sim e^{NS(p)}$ such typical sequences
- the probability of any sequence $\vec{s}$ with empirical distribution $\hat{p}(\vec{s})=p$ is

$$p(\vec{s}) = \prod_{i=1}^{K} p_i^{Np_i}$$

$$= e^{N\sum_{i=1}^{K} p_i \ln p_i}$$

$$= e^{-NS(p)}$$

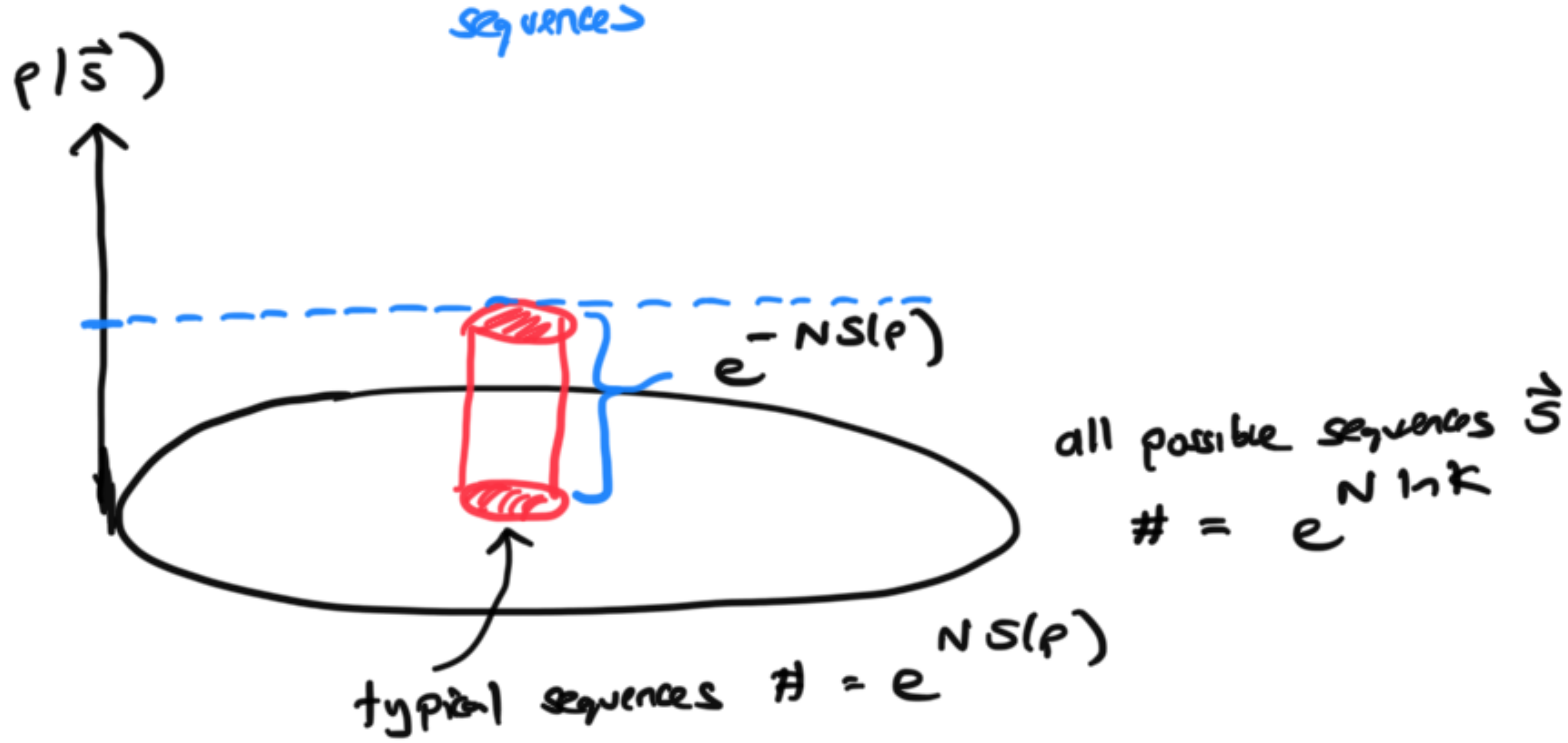
all typical sequences have same

$= e$ probability

- total # of all possible sequences is $K^N = e^{N \ln K}$
- $S(p) < \ln k$ generically $\Rightarrow$ $\underbrace{e^{NS(p)}}_{\#\text{ typical sequences}} \ll \underbrace{e^{N \ln K}}_{\#\text{ all sequences}}$

Picture of distribution $p(\vec{s})$

at large N



$$\boxed{\text{At large } N \text{ almost the entire probability mass of } p(\vec{s}) \text{ is concentrated in on exponentially small fraction of sequence space consisting of } e^{NS(p)} \text{ sequences each with \underline{equal} probability } e^{-NS(p)}}$$

Asymptotic Equipartition Property

- Distribution over microstates $\vec{s}$ induces a distribution over macrostates $\hat{p}$

Analogy

that has a boltzmann form

$$p(\hat{p}) \sim e^{-N D_{KL}[\hat{p} \| p]}$$

$$\sim e^{-\frac{G(\hat{p})}{kT}}$$

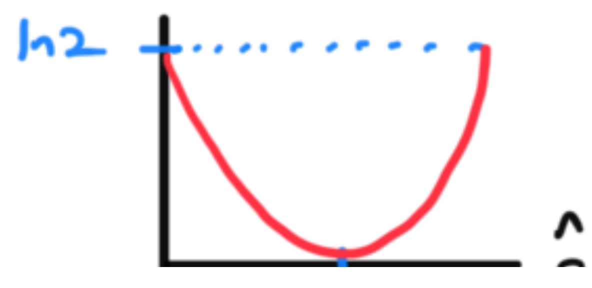
- where "free" energy $G(\hat{p}) = D_{KL}[\hat{p} \| p]$
- temperature $kT = \frac{1}{N}$
- ground state $\hat{p} = p$
- as $N \to \infty$ $T \to 0$
  and distribution of macrostates freezes onto ground state

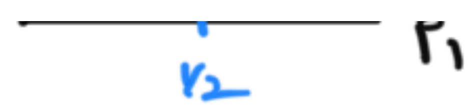
Special case: $p \sim$ uniform

$$p_j = \frac{1}{K} \quad \forall \; j = 1 \ldots K$$

$$D_{KL}[\hat{p} \| p] = \sum_{j=1}^{K} \hat{p}_j \ln \frac{\hat{p}_j}{1/K} = \ln K - S(\hat{p})$$

example : $K = 2$





← uniform empirical distrb is most probable

- **Large deviation theory:** (the above result is a special case of large deviation theory)

Microstate • suppose we have a high dimensional random vector $\vec{s}$ with N components not necessarily iid

Macrostate • suppose we construct a function $F(\vec{s})$

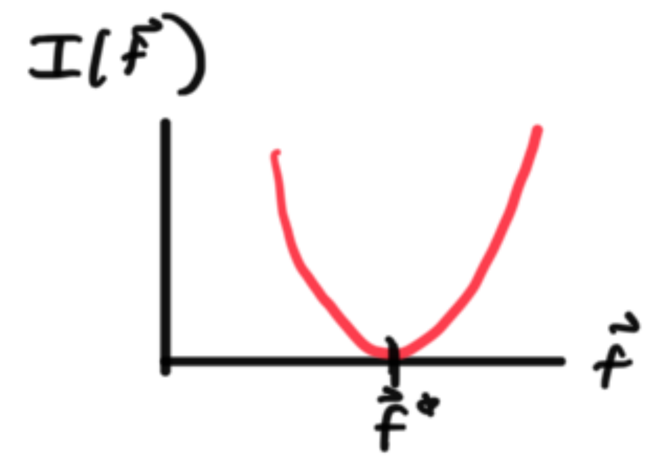
example: $\vec{s} = (x_1, p_1, \dots, x_N, p_N)$ microstate of all molecules of a gas

$\vec{F}(\vec{s})$ = pressure, energy, density etc... of gas

as $N \to \infty$ the probability distribution of the macrostate variable $\vec{F}$

often obeys a large deviation principle:

$$\boxed{\text{Prob}\left[F(\vec{s}) = F\right] \approx e^{-N I(\vec{F})}}$$



· often (but not always): $I(\vec{f})$ is convex

has a unique minimum at $\vec{f}^*$

- so most likely macrostate value is $f = f^*$
- deviations of $\vec{f}$ from $\vec{f}^*$ are exponentially suppressed in N
- $I(\vec{f})$ is exponential rate of suppression

   $=$ large deviation rate function

so far we have seen one example of this

$$\boxed{I(\hat{p}) = D_{KL}[\hat{p} \,\|\, p] = \text{large deviation rate function for } \hat{p}(\vec{s}) \text{ when the distribution is } p}$$

Connection Between Information Theory (KL divergence) and Stat Mech (Gibbs Free Energy)

- suppose we have a physical system with microstates $\vec{s}$ with energies $E(\vec{s})$ in thermal equilibrium at temperature T
- then we know $p(\vec{s}) = \frac{1}{Z} e^{-\beta E(\vec{s})}$ $\beta = \frac{1}{kT}$
- what is the KL divergence between any other distribution $q(\vec{s})$ and the Boltzmann distribution $p(\vec{s})$?

$q(\vec{s})$

$$D_{KL}[q \| p] = \sum_{\vec{s}} q(\vec{s}) \ln \frac{q(\vec{s})}{p(\vec{s})}$$

$$= -\sum_{\vec{s}} q(\vec{s}) \ln p(\vec{s}) + \underbrace{\sum_{\vec{s}} q(\vec{s}) \ln q(\vec{s})}_{-S(q)}$$

$$= -\sum_{\vec{s}} q(\vec{s}) \left[-\beta E(\vec{s}) - \ln Z \right] - S(q)$$

$$= \beta \langle E \rangle_q + \ln Z - S(q)$$

$$\boxed{\langle E \rangle_q = \sum_{\vec{s}} E(\vec{s}) q(\vec{s})}$$

avg energy under q

$$= \beta \left[\langle E \rangle_q + \underbrace{\frac{1}{\beta} \ln Z}_{-F(\beta)} - \frac{1}{\beta} S(q) \right]$$

recall $\boxed{F(\beta) = -\frac{1}{\beta} \ln Z(\beta)}$

free energy of Boltzmann distribution

$$\Rightarrow \boxed{\underbrace{kT}_{\text{thermal energy scale}} \; \underbrace{D_{KL}[q \| p]}_{\text{information theoretic divergence}} = \underbrace{G[q]}_{} - \underbrace{F(\beta)}_{}}$$

$F(\beta)$: Helmholtz free energy of Boltzmann distribution

$$\boxed{G[q] = \langle E \rangle_q - kT\, S(q)}$$

Definition of Gibbs Free energy of a physical system

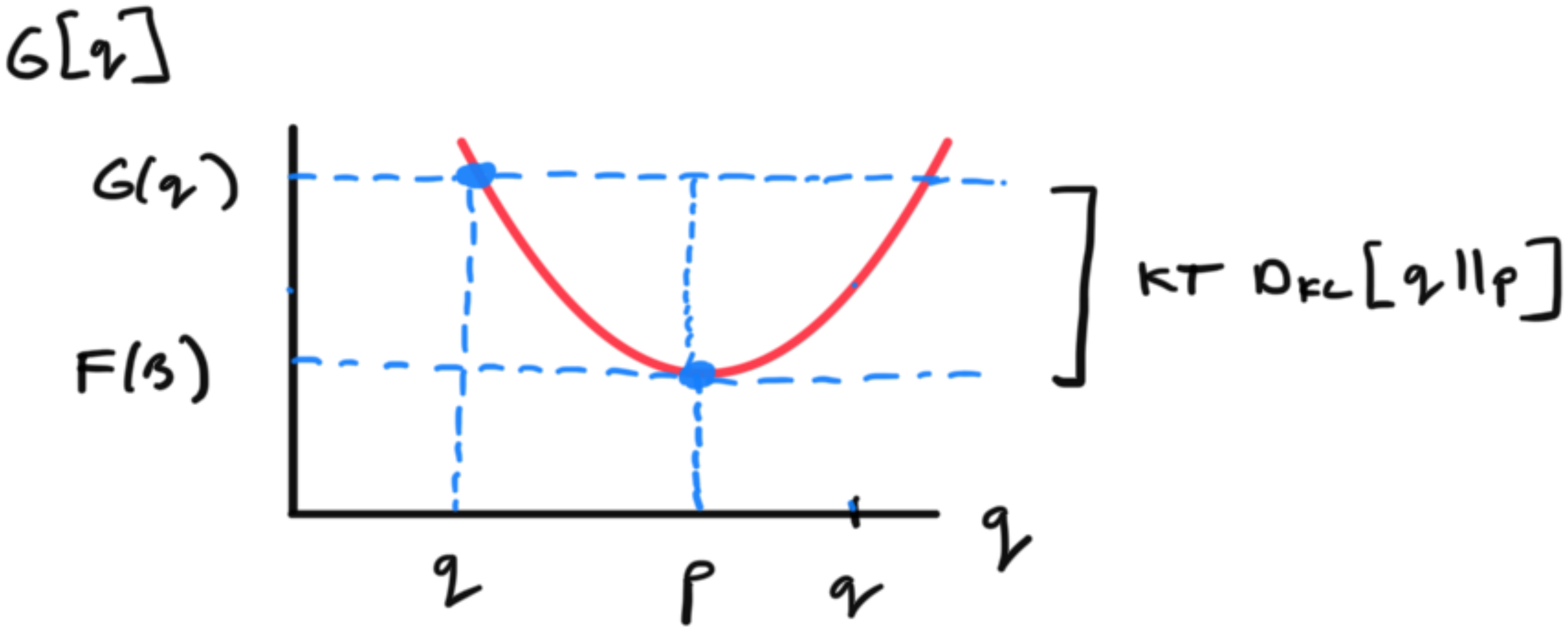
with microstates $\vec{s}$

energies $E(\vec{s})$

but a different probability of microstates $q(\vec{s})$

not necessarily equal to Boltzmann distribution... $p(\vec{s})$

Picture:



Variational characterization of Boltzmann distribution:

Amongst all distributions $q(\vec{s})$, the one that minimizes the

- Gibbs free energy $G[q]$ is the Boltzmann distribution
- the minimum value of the Gibbs free energy is the Helmholtz free energy $F(\beta)$

provides physical intuition:

$$\boxed{G(q) = \underbrace{\langle E \rangle_q}_{\text{minimize energy}} - \underbrace{KT\, S(q)}_{\text{maximize entropy / disorder}}}$$

Competition between energy

Minimize energy

Maximize entropy / disorder

and entropy $\qquad U \qquad$ entropy /

at low temperature kT this is more important

at high temperature kT this is more important

Related variational characterization of Boltzmann distribution: Maximum entropy subject to energy constraint

Consider all possible distributions $q(\vec{s})$

Maximize $S(q)$ subject to $\langle E \rangle_q = E$

$$\boxed{\underbrace{\mathcal{L}[q]}_{\text{Lagrangian}} = \underbrace{-\sum_{\vec{s}} q(\vec{s}) \ln q(\vec{s})}_{S(q)} - \overbrace{\beta}^{\text{Lagrange multiplier}} \overbrace{\left[\sum_{\vec{s}} E(\vec{s}) q(\vec{s}) - E \right]}^{\text{energy constraint}} - \underbrace{\lambda}_{\text{Lagrange multiplier}} \underbrace{\left[\sum_{\vec{s}} q(\vec{s}) - 1 \right]}_{\text{normalization constraint}}}$$

$$\frac{\partial \mathcal{L}}{\partial q(\vec{s})} = 0 \quad \Rightarrow \quad -\ln q(\vec{s}) - 1 - \beta E(\vec{s}) - \lambda = 0$$

$$\frac{\partial \mathcal{L}}{\partial \beta} = 0 \quad \Rightarrow \quad \sum_{\vec{s}} E(\vec{s}) q(\vec{s}) = E$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = 0 \quad \Rightarrow \quad \sum_{\vec{s}} q(\vec{s}) = 1$$

$$-\ln q(\vec{s}) = 1+\lambda + \beta E(\vec{s})$$

$$\boxed{q(\vec{s}) = e^{-(1+\lambda)} \, e^{-\beta E(\vec{s})}}$$

Boltzmann form appears!

Must choose $\beta(E)$ to satisfy energy constraint

must choose $1+\lambda$ to satisfy normalization constraint

Inverse temperature $\beta \longleftrightarrow$ Lagrange multiplier to enforce a given mean energy